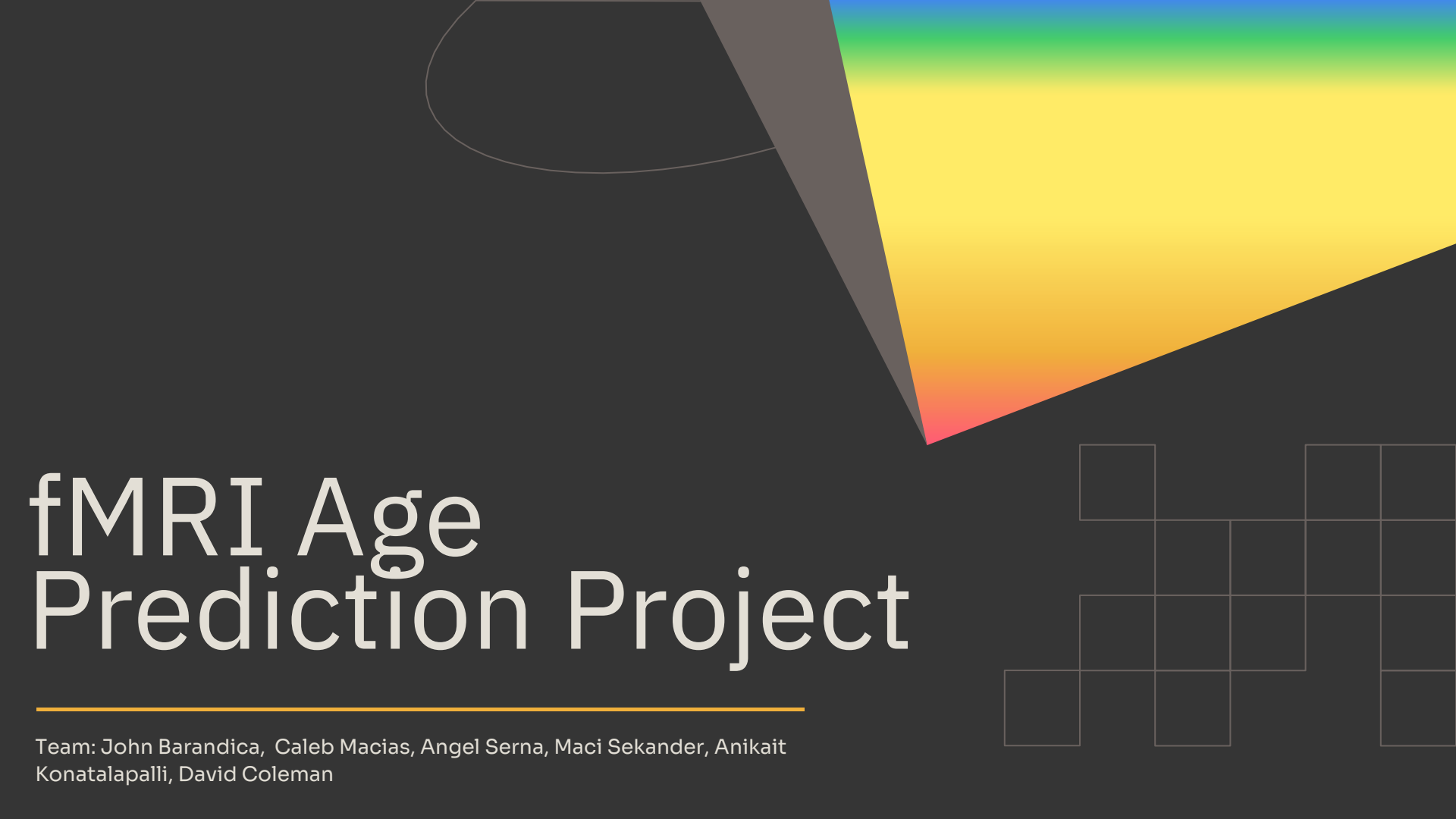




# fMRI Age Prediction Project

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# Introduction

## **Project Title:**

Predicting Age from fMRI Brain Connectivity Data

## **Objective:**

Develop a machine learning model to predict participant age from fMRI connectivity matrices.

## **Why It Matters:**

Brain age prediction supports research in aging, cognitive health, and early detection of neurological issues.

Exploring how brain connectivity changes across the lifespan.

## **Project Focus:**

Applied data preprocessing, dimensionality reduction, and machine learning.

Addressed major challenges: high dimensionality, data imbalance, model instability.

## **Goal:**

Document and present the full trial-and-error journey, from early failures to optimized results.

# Dataset Overview

## **Data Structure:**

**Train and Test folders:** Contain .tsv files for each participant.

Each file holds the upper triangular matrix of brain connectivity values from fMRI scans.

## **Metadata:**

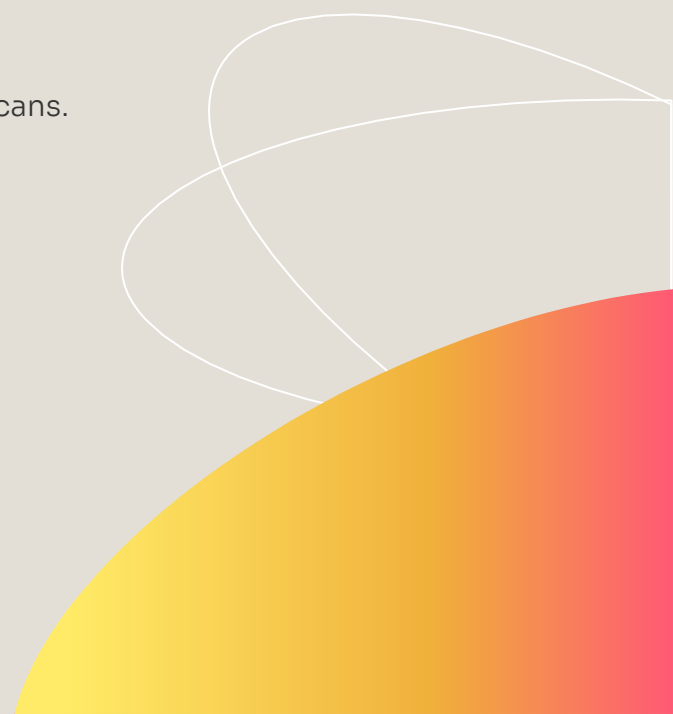
Separate file links participant IDs to age labels and other attributes.

Used to match connectivity data with age for supervised learning.

## **Key Challenges:**

**Data imbalance:** very few participants under age 8 and between 17–21.

High dimensionality: large number of features per participant.



# Initial Pipeline (Baseline Model)

## Data Preprocessing:

Loaded .tsv files and flattened upper triangular matrices into 1D feature arrays for each participant.

Merged feature arrays with metadata (participant ID + age).

## Dimensionality Reduction:

Applied PCA to reduce feature space and minimize overfitting.

Retained principal components explaining most variance.

## Initial Models Tested:

Ridge Regression

Random Forest Regressor

## Outcome:

High error rates, poor generalization.

No predictions for youngest age groups.

# Key Challenges

## Model challenges

1

Data imbalance: Most samples were clustered around teenage to adult ranges

2

Curse of dimensionality: Even after PCA, feature space was large relative to sample size

3

Model instability: Slight parameter changes led to wide fluctuations in output

## How team respond

Realized initial approach was not robust enough

Component alterations and dimensionality tuning

Brainstorming and team collaboration

# Trial and Error - Technique Exploration

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## 1. Data Rebalancing Attempts:

Super-oversampling of underrepresented ages (<8, 17–21).

Helped slightly but led to overfitting.

## 2. Dimensionality Reduction:

Tuned PCA variance thresholds (95%, 90%, 85%).

Best balance found at 90%.

## 3. Data Preprocessing:

Quantile transformation and scaling improved feature distribution.

## 4. Model Experiments:

Ridge Regression (baseline, poor performance)

Random Forest (slightly better, unstable)

XGBoost (most stable and accurate)

CNN (Convolutional Neural Network): used matrices as pseudo-images; struggled with small dataset size

GNN (Graph Neural Network): tried to model brain region connectivity as graphs; limited by dataset size and structure

# Synthetic Data Attempt

**Idea:** Generate synthetic samples for missing age ranges

**Method:**

Created pseudo-connectivity matrices and paired with missing age labels

**Process:**

- Compilation:
  - compiled all patient Metadata and fmri scans into one main csv file
- Split:
  - Separated patients based upon ages 5-5.5, 5.51-6 and so on
- Range
  - For each age group we took the range for each individual's numerical features (ie. Age, BMI, p\_factor\_fs,)
- Generation:
  - Based on these ranges randomly selected values and generated data for new individual patients in each under represented age range



We generated synthetic samples by adding noise to existing participant matrices to fill gaps in underrepresented age ranges (<8, 17-21). The approach caused unstable training and unrealistic patterns, so it was abandoned.

# Synthetic Data Attempt

## **Result:**

Increased file size dramatically

Crashed multiple training runs

Introduced artificial patterns → worsened generalization

**Decision:** abandoned synthetic data and moved to real data

# The Breakthrough: Alternative Solutions

## **Sample Weighting in XGBoost:**

Assigned higher weights to underrepresented age groups

Model became more balanced and stable

## **Data preprocessing improvements:**

Applied quantile transformation to reduce skewness

Standard scaling post PCA

## **Hyperparameter tuning:**

Grid search + cross-validation for optimal XGBoost settings

Clear improvement in model reliability and interpretability



# Model Evaluation

## **Metrics Used:**

Root Mean Squared Error (RMSE) to measure prediction accuracy.

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$R^2$  Score to assess variance explained by the model.

## **Results:**

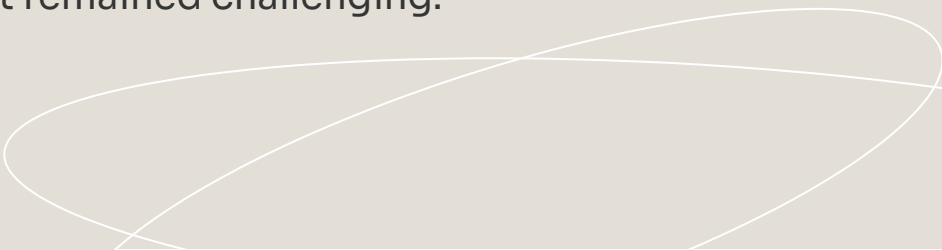
Early models (Ridge, Random Forest) had high RMSE and poor performance on underrepresented ages.

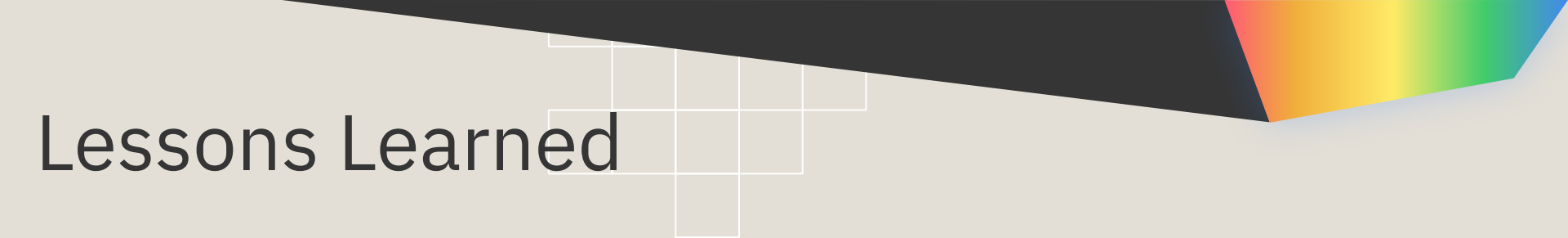
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Final pipeline (PCA + XGBoost + sample weighting) delivered the most consistent and balanced results across all age groups.

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Predictions for rare ages slightly improved but remained challenging.





# Lessons Learned

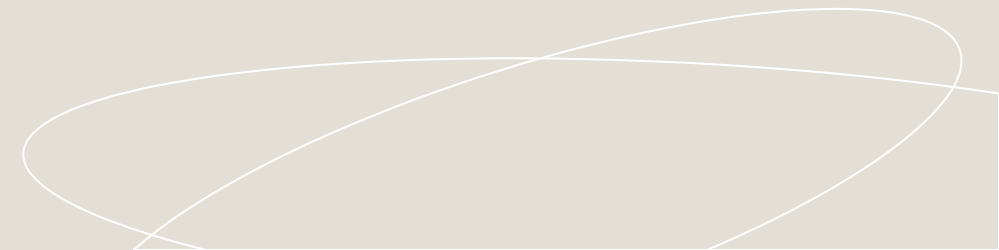
## Key Takeaways:

**Trial and error was essential:** Many strategies failed before the final model succeeded.

**Synthetic data is risky:** Introduced noise and instability rather than solving imbalance.

Sample weighting worked better: Prioritizing underrepresented age groups improved performance.

Dataset limitations shaped outcomes: Small sample size and severe class imbalance restricted the power of deep learning methods like CNN and GNN.



# Next Steps & Future Plans

## **Future Improvements:**

Explore advanced architectures:

Graph Neural Networks (GNN) to better model brain connectivity as networks.

Convolutional Neural Networks (CNN) with larger or more structured datasets.

**Data acquisition:** Seek more balanced datasets, especially ages <8 and 17–21.

Refine preprocessing: Experiment with new feature extraction or graph-based representations.

**Long-term goal:** Use the model to predict biological brain age for clinical screening and research.

# Conclusion

## **Summary of Work:**

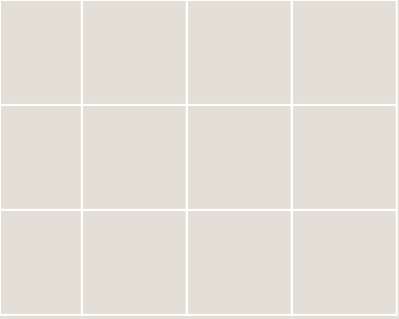
Developed and refined a machine learning pipeline to predict age from fMRI connectivity data.

Faced and overcame major challenges through iterative testing and model tuning.

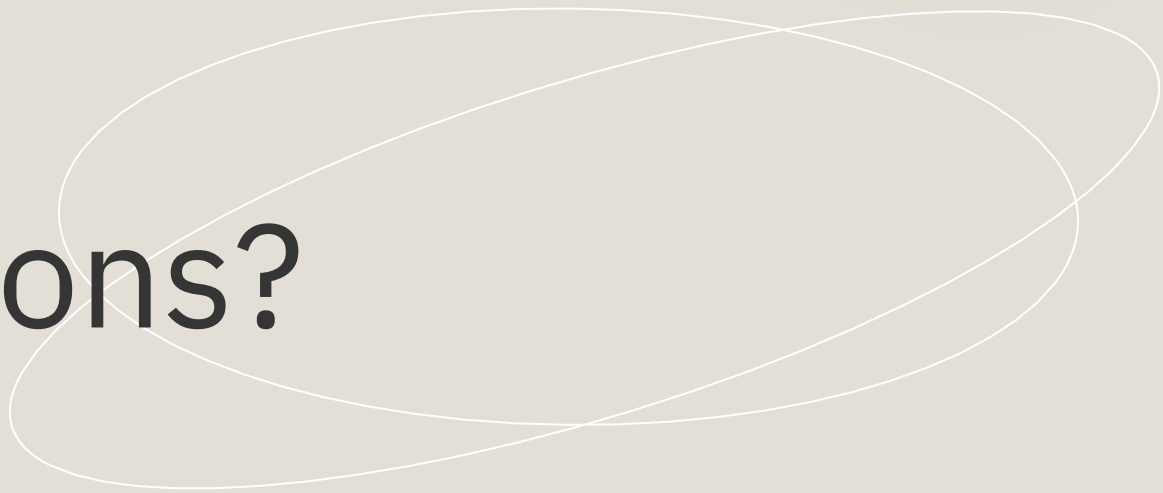
Final model (PCA + XGBoost + sample weighting) delivered the best balance of stability and accuracy.

## **Final Thought:**

The project reinforced that in real-world data science, persistence, experimentation, and adaptability are just as important as technical skill.



Any questions?  
Ask away!





Thank you